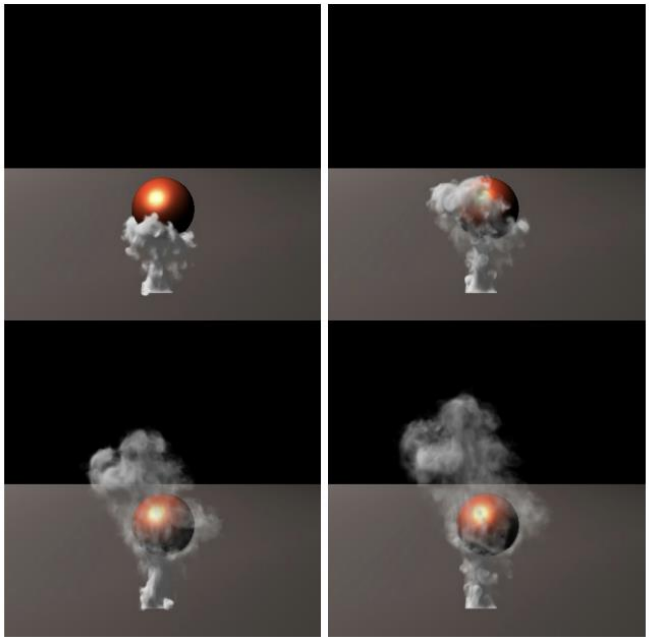


Figure 1. A set of sequential figures showing the smoke rising around a sphere. The simulation was done on a $256 \times 256 \times 256$ grid up-sampled from a $128 \times 128 \times 128$ coarse grid using 8192 particles based on the techniques proposed in this paper. Ray-marching was used in this simulation.

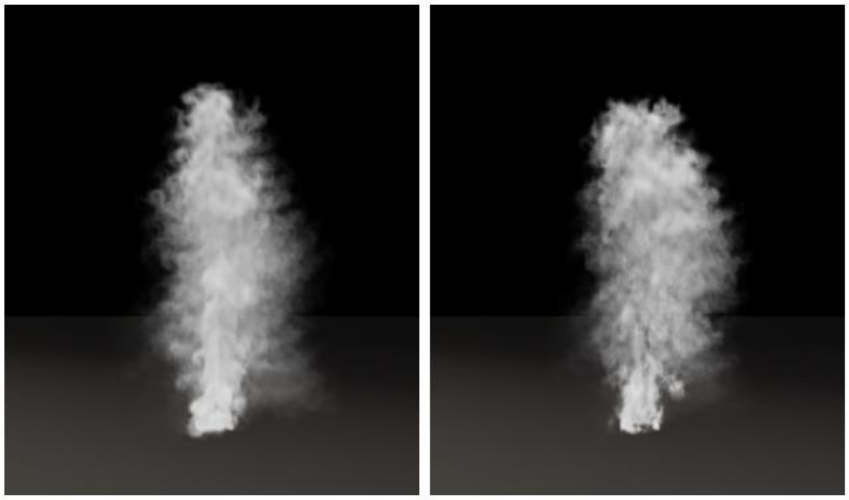


Figure 2. A comparison of using the spatial adaptive vorticity confinement (*Left*) and the vorticity confinement in [6] (*Right*) as the vorticity forcing method. We clamp the vorticity magnitude to a high range for both cases.

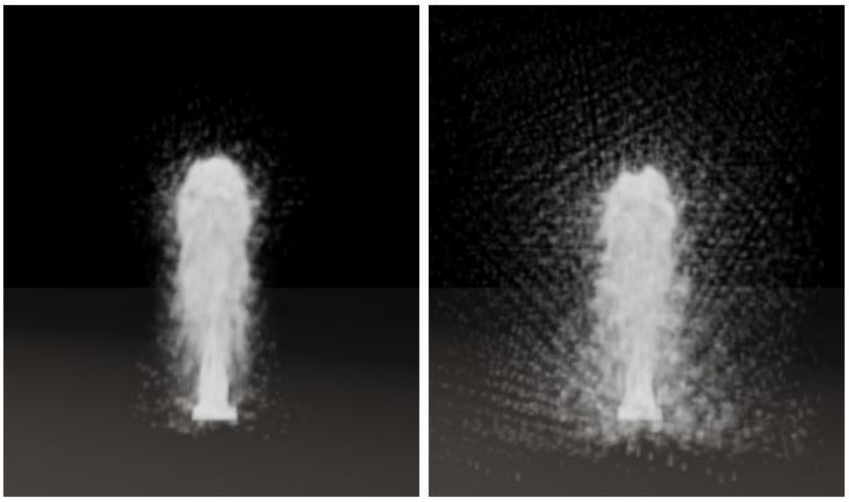


Figure 4. A comparison of using our efficient adaptive particle handling (*Left*) and without using it (*Right*). Each activated particle brings the density to show its state. Buoyancy force was ignored in this simulation.

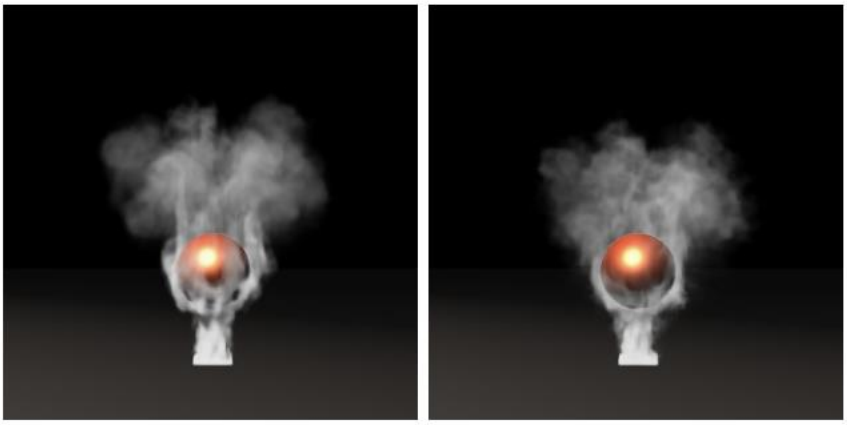
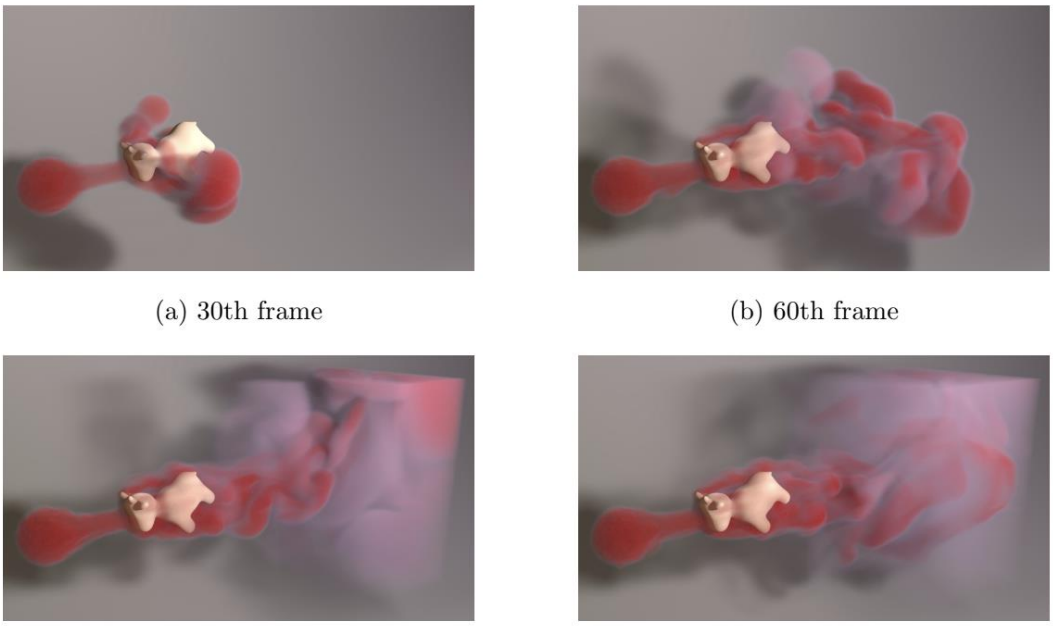


Figure 5. Smoke rising around a sphere shows the difference of using our efficient adaptive particle handling (*Left*) and without using it (*Right*). This example is a $128 \times 128 \times 128$ simulation with a $64 \times 64 \times 64$ coarse grid.

Smoke Plume and Solid. Stable fluids also works smoothly in 3-Dimension. Here we show a smoke plume interacting with a solid cartoon cow object [10] fixed at some world positions during the simulation process. A slight perturbation is added to a spherical smoke source, and it is enforced to continuously emit smoke in some fixed directions within each timestep. After colliding with the solid cow and the fluid domain boundaries, it produces the visual effects shown in Figure 3.5. The simulation runs in $128 \times 256 \times 128$ resolution, and the density field advected along with velocity is stored for visualization and for rendering.



(a) 30th frame

(b) 60th frame

(c) 90th frame

(d) 120th frame

Figure 3.5: 3D Smoke Plume and Solid [Density]: Sable Fluids

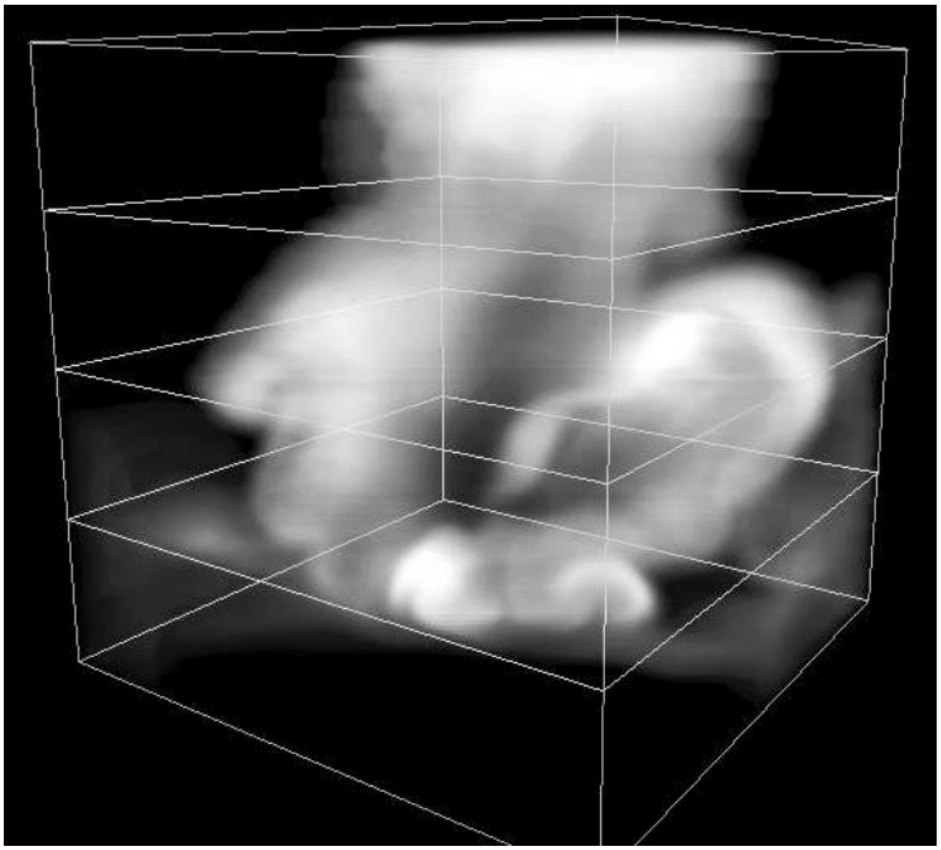


Figure 3.1: Volume rendered with our distributed rendering algorithm. The wire cubes show one part of the grid which is simulated on one node of the cluster.

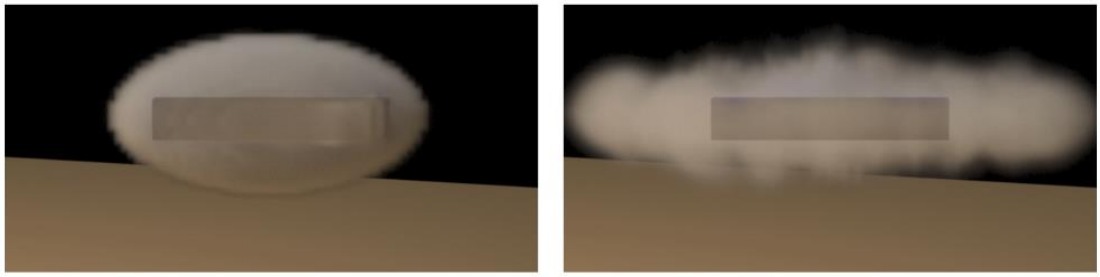


Figure 6: Two stills from the rotor animation. A box is rotating inside the smoke cloud causing it to disperse. Notice how the smoke is sucked in vertically towards the box as it is pushed outwards horizontally. The simulation time for a 120x60x120 grid was roughly 60 seconds/frame.



Figure 3: Rising smoke. Notice how the vorticies are preserved in the smoke. The simulation time for a 100x100x40 grid was roughly 30 seconds/frame.

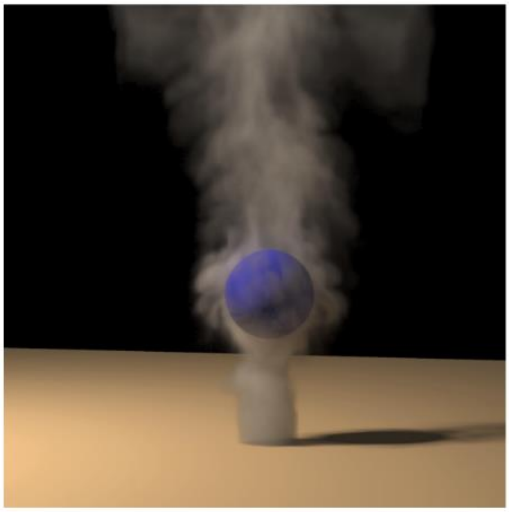


Figure 5: Rising smoke swirling around a sphere. Notice how the smoke correctly moves around the sphere. The simulation time for a 90x135x90 grid was roughly 75 seconds/frame.

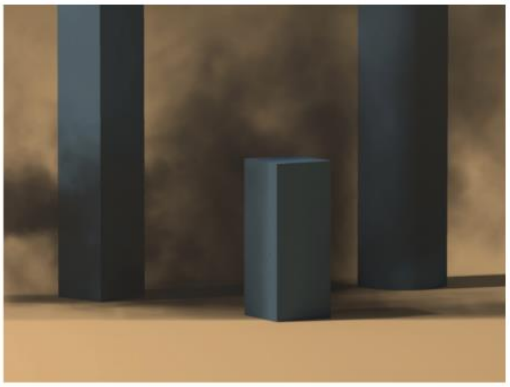


Figure 4: Low albedo smoke passing through several objects. Each object interacts with the smoke and causes local turbulence and vorticity. The simulation time for a 160x80x80 grid was roughly 75 seconds/frame.



Figure 7: Six frames rendered using our interactive hardware renderer of the smoke. The simulation time for a 40x40x40 grid was roughly 1 second/frame.

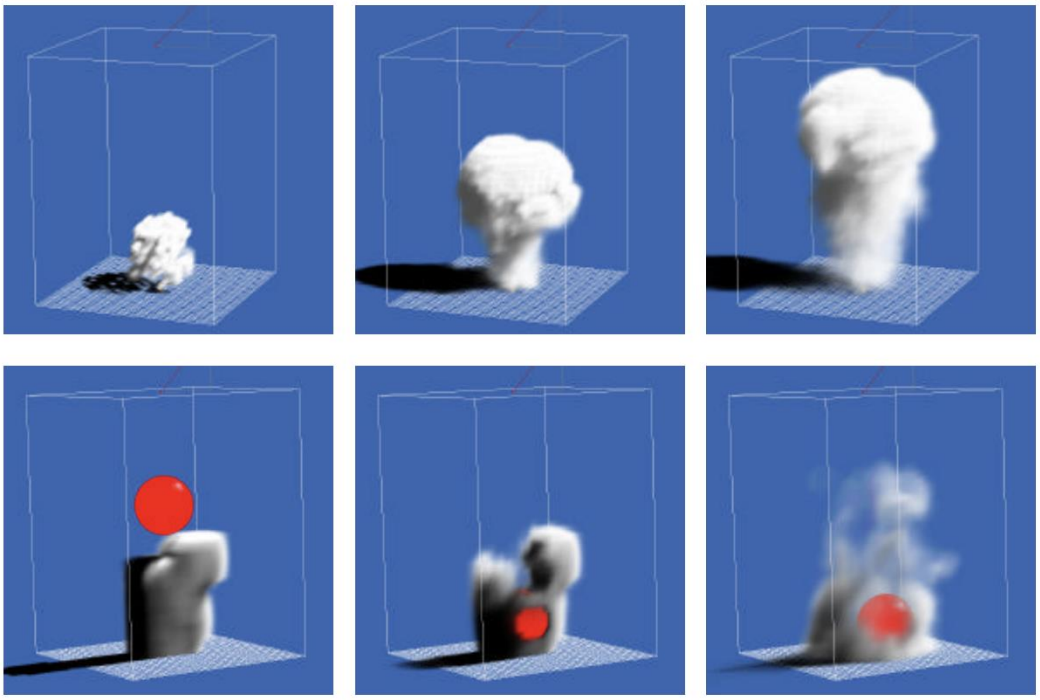


Figure 8: Snapshots from our prototypes: two-dimensional solver (top) and three-dimensional solver (below). In each case densities and forces were added interactively by the user.